

Practice set 4

Merit and Excellence problem solving

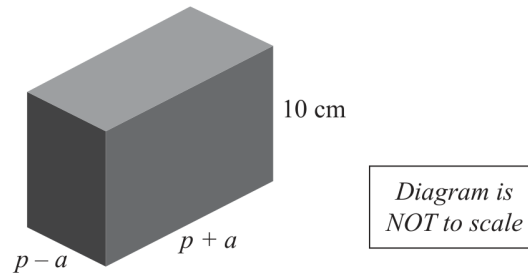
Write in the spaces

- (d) Consider the following two curves:
 $x^2 = y^2 + 1$ and $y = (x - 1)(x + 1) - 2$

Find the co-ordinates of each intersection point of the two curves.

- (d) Junyang realises that the **surface area** of the cuboid will not be the same as the surface area of the cube unless he also changes the height. He decides to make the height of the cuboid 10 cm.

He wants to find out which value of p would result in the cube having the same surface area as the cuboid. To do this, he needs to form and solve an equation for p .



- (i) If the surface area of the cube is the same as the surface area of the cuboid, show that $2p^2 - 20p + a^2 = 0$.

Remember that the surface area of the cube is $6p^2$ cm².

- (ii) As mentioned in part (i), if the surface area of the cube is the same as the surface area of the cuboid, then $2p^2 - 20p + a^2 = 0$.

By using the discriminant (Δ), find the largest possible **whole** number value that a could take in this context.

Use this value of a to find the dimensions of both the cube and the cuboid.

Explain your reasoning clearly.

- (c) A calendar can be presented in the following way, where each day is given a number from 1 to 365. This is the beginning of a year's calendar:

M	T	W	TH	F	SA	SU
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31
32	33	34	35	36	37	38
39	40	41	42	43	44	45
46	47	48	49	50	51	52
53	54	55	56	57	58	59
60	61	62	63	64	65	66
67	68	69	70	71	72	73

Jo draws a 4-by-4 square on the calendar to check a claim that she heard:

“the sums of the diagonally opposite corners are always the same, no matter where you make your square”. In other words, when you add the numbers in the orange corners, it is the same as when you add the numbers in the blue corners.

Jo wonders if the claim will still be true no matter where she starts the square, so she begins an investigation using algebra:

A			

- (i) Use algebra to prove that, no matter where the 4-by-4 square is drawn on the calendar, **the sum of the orange corners must be the same as the sum of the blue corners.**

- (ii) Jo wonders about the **products** of the diagonally opposite corners: could they be the same?

Use algebra to prove that it is **not** possible to draw any 4-by-4 square for which the products of the diagonally opposite corners are the same.

(c) Solve $\sqrt{2x+3} = 3x$.

(d) Suppose $Q(x) = fx^2 + gx + h$, where f , g , and h are constants. The “reciprocal polynomial” of $Q(x)$ is defined as $Q^*(x) = hx^2 + gx + f$, where the coefficients are in the reverse order.

(i) Find the solutions of the equation $Q(x) = Q^*(x)$.

- The roots of $Q^*(x) = 0$ are multiples of A and of B, i.e. the roots are kA and kB for some constant k .